Foundation Models for Semantic Segmentation of Thick/Thin Clouds and Cloud-shadows: A Comparative Study

Calvin Nguyen
North Carolina State University
Raleigh, NC, USA
cvnguye3@ncsu.edu

Ranga Raju Vatsavai
North Carolina State University
Raleigh, NC, USA
rrvatsav@ncsu.edu

Abstract

Clouds pose the most significant hindrance to satellite imagery analysis. A crucial step in pre-processing satellite imagery involves masking these clouds, which ensures reliable downstream geospatial analysis. However, this masking also reduces the effectively analyzed region. To address this, recent studies have explored imputing missing values under clouds to restore a complete image. Unfortunately, the inaccuracy of the off-the-shelf masks (e.g., QA band from Landsat 8) adversely affects downstream geospatial analysis and machine learning tasks. Accurate imputation necessitates precise detection of clouds, including thin clouds and shadows. Of recent cloud segmentation models segmenting a subset of the cloud phenomena (i.e., thick clouds, thin clouds, and cloud-shadows), their accuracy, though promising, remains insufficient; even fewer models segment all three phenomena with adequate performance. Foundational models show promise for image segmentation tasks, but studies with them in cloud segmentation are sparse. In this paper, we present a thorough experimental analysis, studying different properties of foundational model and their effectiveness on this comprehensive task at different levels of transfer learning. This analysis will explore foundational model architecture, pretraining dataset, and scheme. The main result we found is the transformer architecture demonstrates superior performance on granular details on boundaries and small-sized segments, compared to convolutional or hybrid architectures.

CCS Concepts

• **Computing methodologies** → Supervised learning; *Transfer learning*; *Neural networks*; **Image segmentation**; Unsupervised learning; • **Applied computing** → *Environmental sciences.*

Keywords

Cloud Segmentation, Foundational Model, Remote Sensing, Self-supervised, Landsat 8

ACM Reference Format:
Calvin Nguyen and Ranga Raju Vatsavai. 2025. Foundation Models for Semantic Segmentation of Thick/Thin Clouds and Cloud-shadows: A Comparative Study. In *The 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25), November 3–6, 2025, Minneapolis, MN, USA.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3748636.3762766

SIGSPATIAL '25, Minneapolis, MN, USA

ACM ISBN 979-8-4007-2086-4/2025/11
https://doi.org/10.1145/3748636.3762766

1 Introduction

Clouds and cloud shadows significantly impede downstream machine learning tasks, reducing the performance of applications such as land use land cover (LULC) mapping, soil erosion and runoff, and fire scar detection. Such analyses require cloud-free imagery for accurate predictions. The primary obstructions in satellite images are clouds and related phenomena (i.e., thin clouds and cloud shadows). Practitioners have two main options for addressing this phenomenon: either masking out or imputing (replacing with predicted values) the clouds. However, both approaches first require accurate identification of clouds and cloud shadows in the satellite image, which is a task of image segmentation. For full-image analysis, neural networks represent the state-of-the-art models in the geospatial domain. Neural networks outperform traditional computer vision algorithms due to their highly parameterized modules like convolutions and attention mechanisms, making them top performers in benchmark datasets.

To thoroughly test these cloud detection algorithms, the USGS published the most widely cited for Landsat 8 cloud detection, L8 Biome [6]; a dataset still to this day unresolved for harder cloud-related phenomena of thin clouds and cloud-shadows. Many algorithms segment a subset of atmospheric phenomena, but few address all three on the L8 Biome dataset. While reducing thin clouds to clouds or cloud-shadows to the background for masking is acceptable, merging these classes disallow targeted downstream imputation. Despite the limitations of existing machine learning models in accurately detecting these three atmospheric phenomena, recent advances in general-purpose foundational models show significant promise for improving downstream tasks; however, studies directly comparing them for semantic segmentation remain limited.

Contributions: (i) We analyze how foundation model architecture, pretraining data, pretraining scheme, and promptability influence transfer learning performance for segmenting clouds, thin clouds, and cloud-shadows on the L8 Biome dataset, (ii) We evaluate performance on challenging cases, segmentation of thin clouds and cloud shadows and over the snow/ice biome.

2 Literature Review

General-Purpose Models: Deep neural networks (CNNs or transformers) lead image-based cloud segmentation. Foundation models are general-purpose neural networks designed for task adaptability via *transfer learning*, acting as feature encoders; some are adapted for geospatial data by their pretraining dataset. The Segment Anything

Model (SAM) [10] introduced promptability, allowing for some potential few-shot learning capabilities. These powerful models and their diverse design decisions show promise for cloud segmentation.

Purpose-built Models: Neural network solutions specifically for L8 Biome cloud segmentation leverage convolutions and attention. Most address only subsets (e.g., cloud-only, cloud/shadow, or cloud/thin-cloud). These approaches emphasize scale, novel modules, or multi-image setups [4, 11, 16, 18]. Few models, like SCTNet [13], tackle all cloud phenomena; their architectures vary from pure convolutions to transformers or hybrids.

While detecting (thick) clouds is largely solved problem (98%-99% accuracy), detecting thin clouds and cloud shadows present a significantly harder task, even more so simultaneously. Thin clouds and cloud shadows often appear as background over snow/ice biomes, necessitating explicit distinction in performance analysis. Researchers addressing only one of these challenging classes alongside clouds often yield suboptimal models for downstream imputation. Recent foundational models employ diverse design decisions, from the architecture to the pretraining dataset and scheme. They often show similar performance on their respective subsets of cloud phenomena, making it unclear how to best choose/design a foundational model for the full range of cloud phenomena.

Table 1: Properties of foundational models.

Model	Architecture	Pretraining Dataset	Pretraining Scheme
HQ-SAM [9]	ViT	SA-1B, HQSeg-44K	Supervised (+prompting)
Prithvi [8]	ViT	HLS (Sentinel-2/Landsat 8)	Self-supervised (MAE [7])
SatlasNet [2]	Swin	SatlasPretrain	Supervised (many tasks)
LSKNet [12]	CNN	ImageNet-1K	Supervised

3 Methodology

For a precise description, we want a model that outputs a 224 x 224-sized segmentation map of the following classes: clear, (thick) cloud, thin cloud, and cloud-shadow. Thick clouds are opaque clouds that obscure the land underneath. Thin clouds are classified as transparent clouds or pixels with unclear determination as a cloud. Cloud-shadows are shadows that are derived from soley clouds or thin clouds. The background is composed of everything else in the imagery; the class does admit clouds if the cloud opacity is weak enough (50% or lower) to leave downstream analysis largely undisturbed.

We use a variety of models to explore how different architectures, pretraining datasets, and pretraining schemes can handle cloud segmentation. We will look at a traditional algorithm (FMask 3.3), a promptable model (HQ-SAM), two geospatial ViT models (Prithvi and SatlasNet), and a geospatial convolutional model (LSKNet).

FMask in our studies serves as the baseline model for realistic cloud segmentation for it's usage in the QA band of Landsat 8 imagery. We use HQ-SAM over the original SAM because it ameliorates SAM's pitfall of making filled predictions. For Prithvi, SatlasNet, and LSKNet, we chose representative, open-weight models that were designed for Landsat 8 imagery, with either in-domain (geospatial) and out-domain datasets, and different flavors of supervision with the most popular self-supervised method (MAE).

3.1 Foundational Model Specification

For foundational models, we describe their attributes and common choices. Table 1 summarizes which models adopt which choices. The structure of a foundational model strongly impacts task performance. Vision architectures range from global to local receptive fields. Vision transformers [5] compute attention across all patches, while convolutional networks extract local patterns. Hybrid designs like Swin [14] restrict attention within windows, and others add prompt encoders to incorporate partial labels. Pretraining data shapes model biases, particularly across domains (e.g., geospatial). ImageNet [3] remains the most common non-geospatial dataset. SA-1B [10] enabled SAM's prompting. For geospatial tasks, SatlasPretrain [2] was released by the Allen Institute, while IBM and NASA harmonized Sentinel-2 and Landsat-8 imagery for Prithvi. Architecture and data are linked by the pretraining scheme. A standard choice is ImageNet supervised classification. Others combine tasks such as classification, segmentation, and regression, as with SatlasNet. On the self-supervised side, masked autoencoders dominate, learning strong representations by reconstructing masked images.

3.2 Model Adaptation for Segmentation

In this subsection, we outline the training methodology for the foundation models; the models in this paper are adapted for segmentation by: (i) transfer learning, where pre-trained weights are repurposed for new tasks or (ii) prompt engineering, submitting partial labels with no amount of retraining.

The testing procedure of the trained models is a simple holdout with 60% of the data for training, 15% for validating, and 25% for testing. We will do a full comparison of the trainable models at 3 levels of *transfer learning*. Initially, only the task head of the model is trained (i.e., *feature extraction*). Taking those weights, half the encoder is trained (i.e., *half-frozen*). Finally, the previous weights are trained with the entire model fully unfrozen (i.e., *fine-tuning*). Following the training procedure, we elaborate on the model training/evaluating decisions when using the models.

For the baseline model, FMask 3.3, we make two key adaptation choices. We turn off buffering parameters to finely target classes and do full-scene processing to not hamper the cloud-shadow matching process.

HQ-SAM does not inherently segment for semantics. To add semantics, each class is prompted with n points and another n points for the rest of the classes; the highest value logit among the classes determines the output. For single-class images, this procedure is not possible, therefore we assign the whole image as the class. Because of direct access to ground truth, HQ-SAM cannot be reasonably compared with the baseline model or trained foundational models. It should be treated as a glimpse into the aspect of promptability of newer foundational models for cloud segmentation.

Some models take more than just *red, green, and blue* (RGB) bands, which requires some usage decisions. We restrict the foundational models to handle only RGB bands as other cloud segmentation models [4, 16, 18] have demonstrated comparable performance when using just RGB bands. To focus on how foundational models capture spatial information, we simply duplicate RGB bands in-place of other bands. On the other hand, FMask requires *all 11 Landsat 8 bands* because of the rule-base nature. For the output side of the foundational models, we use the author recommended task heads to stay true to their design.

4 Experiments

4.1 Data Processing

The Landsat 8 Biome (*L8 Biome*) dataset[1] serves as the benchmark for segmenting clouds, thin clouds, and cloud shadows. We selected this dataset for three key reasons: (1) it is the most widely cited benchmark for cloud segmentation, with numerous published solutions; (2) performance on thin clouds and cloud shadows remains subpar; and (3) the dataset is categorized by biome.

We focus purely on Landsat 8 imagery for it's coarse 30 m resolution compared to other optical satellites like Sentinel-2 (10 m) and Planet (1 m). 30 m resolution Landsat 8 represents the more difficult resolution to segment over because of the non-smooth behavior of coarse digitizing of objects [1].

Non-overlapping 224×224 tiles are taken from the scenes, excluding nodata-containing images and randomly sampling 25% (~18,000 images) for computational efficiency. Though the dataset size may not be sufficient for training a deep learning model from scratch, this size suffices for transfer learning on our 4-class image segmentation problem.

4.2 Performance Evaluation Metrics

For performance evaluation, accuracy, Intersection-over-Union (per-class and macro IoU), and F1-measure (per-class and macro F1) will be used. Beyond main segments, IoU will uniquely evaluate the boundaries and small objects for each class.

HQ-SAM will be tested for its prompting performance using an exponentially increasing number of single-point prompts. Specifically, 1, 10, 100, and 1,000 pairs of foreground and background pairs of points per-class will be used, totaling 8, 80, 800, and 8,000 prompt points depending on image capacity.

4.3 Transfer Learning Setup

To enable the evaluation, we perform all transfer learning phases (refer to subsection 3.2) with the following setup for Prithvi, SatlasNet, and LSKNet. We employ AdamW optimization [15] with default parameters besides a learning rate of 5×10^{-5}. The optimizer minimizes the cross-entropy loss for the trained models besides for LSKNet which minimizes the author recommended UNetFormer [17] hybrid cross entropy and dice loss (on the auxiliary task head). To address severe class imbalance, we weight the loss inversely quadratic to class frequency, emphasizing rare classes. We implement a 1-epoch learning rate warm-up at 10% of the target rate to facilitate task adaptation. Training uses a batch size of 32 images, resorting to reduced power-of-2 sizes with gradient accumulation when VRAM constraints occur. Models are trained until convergence, defined as five consecutive epochs with 0.001 difference in validation loss to best epoch.

[1]Data can be downloaded from: https://landsat.usgs.gov/landsat-8-cloud-cover-assessment-validation-data.

Table 2: Trained foundational model comparison.

Model	Class	Feature Extraction			Half-frozen			Fine-tuning		
		Acc	IoU	F1	Acc	IoU	F1	Acc	IoU	F1
Prithvi	Clear	–	0.88	0.93	–	0.92	0.96	–	0.90	0.95
	Cloud	–	0.86	0.93	–	0.91	0.95	–	0.89	0.94
	Thin Cloud	–	0.58	0.73	–	0.66	0.80	–	0.64	0.78
	Cloud-shadow	–	0.37	0.54	–	0.46	0.63	–	0.44	0.61
	Overall	0.89	0.67	0.78	0.93	0.74	0.83	0.91	0.72	0.82
SatlasNet	Clear	–	0.83	0.91	–	0.87	0.93	–	0.89	0.94
	Cloud	–	0.82	0.90	–	0.87	0.93	–	0.87	0.93
	Thin Cloud	–	0.49	0.65	–	0.58	0.73	–	0.65	0.79
	Cloud-shadow	–	0.00	0.00	–	0.27	0.43	–	0.45	0.62
	Overall	0.86	0.53	0.62	0.89	0.65	0.75	0.91	0.72	0.82
LSKNet	Clear	–	0.89	0.94	–	0.90	0.95	–	0.91	0.95
	Cloud	–	0.87	0.93	–	0.90	0.95	–	0.91	0.95
	Thin Cloud	–	0.59	0.74	–	0.64	0.78	–	0.66	0.79
	Cloud-shadow	–	0.32	0.48	–	0.41	0.58	–	0.42	0.59
	Overall	0.89	0.66	0.77	0.91	0.71	0.81	0.92	0.72	0.82

Table 3: Combined comparison of overall metrics, boundary IoU at boundary buffering radii $r = 0$ and $r = 2$, and IoU for objects of maximum sizes 5, 50, and 500 pixels.

Model	Overall Metrics			Boundary IoU		IoU by Object Size		
	Acc	IoU	F1	$r = 0$	$r = 2$	5 px	50 px	500 px
FMask	0.71	0.36	0.45	–	–	–	–	–
HQ-SAM (10 Pairs)	0.87	0.62	0.73	–	–	–	–	–
Prithvi	0.91	0.72	0.82	0.27	0.45	0.20	0.31	0.47
SatlasNet	0.91	0.72	0.82	0.27	0.47	0.16	0.27	0.44
LSKNet	0.92	0.72	0.82	0.21	0.41	0.15	0.23	0.40

5 Results

Now with the methodology and experiments elaborated[2], we will walk through our results. We start with the results of individual model performance, onto the final comparisons for overall performance, then end with performance on various key issues with cloud segmentation: segmenting boundaries, small objects, and over the snow/ice biome.

(1) Foundational model architecture plays an insignificant role on overall segment performance (table 3), but (2) sports a significant performance when handling granular phenomena (i.e., boundaries and small objects). Prithvi demonstrated the best performance on granular phenomena, next with SatlasNet (with it's hybrid architecture), then LSKNet trailing with it's pure convolutional features. (3) Pretraining dataset makes no difference in overall fine-tuned performance, unless (4) during low-training transfer learning with pre-training scheme that ignores shadow features (i.e., SatlasNet). Prithvi achieves a slight advantage over LSKNet and a major advantage over SatlasNet on segmentation over the snow biome (table 4); (5) pretraining data and scheme may play a role in SatlasNet's poor fine-tuned performance on the snow-ice biome.

Promptability could be a key technique for cloud segmentation, as it effectively segments clouds and the clear class (table 5); however, the harder phenomena—thin clouds and cloud shadows—still require further development. Promptable foundational models could greatly benefit from pretraining on remote sensing imagery for it's no-fine-tuning applications.

[2]Source code is publicly available at: https://github.com/cvnguye3-ncsu/l8-biome.

Table 4: Comparison of class-wise and overall performance for fine-tuned foundational models on the Snow/Ice biome.

Class	Prithvi			SatlasNet			LskNet		
	Acc	IoU	F1	Acc	IoU	F1	Acc	IoU	F1
Clear	–	**0.86**	**0.92**	–	0.67	0.80	–	0.85	**0.92**
Cloud	–	0.55	0.71	–	0.27	0.42	–	**0.66**	**0.79**
Thin Cloud	–	0.70	0.82	–	0.60	0.75	–	**0.73**	**0.84**
Cloud-shadow	–	**0.47**	**0.64**	–	0.36	0.53	–	0.33	0.50
Overall	**0.87**	**0.65**	**0.77**	0.73	0.47	0.63	**0.87**	0.64	0.76

Table 5: Comparison of performance metrics for HQ-SAM for n pairs of foreground and background prompt points for each class, meaning at most $8n$ total number of data points were selected.

Class	1 Pair			10 Pairs			100 Pairs			1,000 Pairs		
	Acc	IoU	F1	Acc	IoU	F1	Acc	IoU	F1	Acc	IoU	F1
Clear	–	0.75	0.86	–	0.86	0.93	–	0.86	0.93	–	0.87	0.93
Cloud	–	0.69	0.82	–	0.84	0.91	–	0.83	0.91	–	0.84	0.91
Thin Cloud	–	0.36	0.53	–	0.54	0.70	–	0.54	0.70	–	0.55	0.71
Cloud-shadow	–	0.15	0.26	–	0.25	0.40	–	0.29	0.45	–	0.31	0.47
Overall	0.78	0.49	0.62	0.87	0.62	0.73	0.87	0.63	0.75	0.88	0.64	0.76

6 Conclusion

We've detailed the foundational model architecture, pretraining dataset, and training scheme, showing their impact on cloud segmentation. Other factors, like model size and multi-scale feature extraction, are also important. While larger models often perform better, there's a trade-off with computational resources and reliability. The core of a foundation model lies in its encoder (the feature extractor), not the task-specific head. However, direct comparison is difficult because models have incompatible feature shapes and semantics. Some architectures produce features at different scales, while others use prompt encoders or convolutional features. A unified segmentation head would allow for a consistent comparative analysis. In our dataset, segmenting coarser-resolution imagery is complicated by the non-smooth digitization of objects. Landsat 8 (30 m) likely represents a performance lower bound compared to higher-resolution data. Exploring Sentinel-2 (10 m) or Planet (1 m) imagery might improve results due to higher spatial resolution. We hope these findings help researchers use foundational models for cloud segmentation, especially for challenging classes like thin clouds and cloud shadows, and other complex spatial phenomena.

Generative AI Usage Disclosure: During the preparation of this work, we used Generative AI tools (e.g., Google Gemini) for language editing tasks to improve the grammar and clarity of the authors' written text; we did not AI generate any content in this paper. All AI-edited text was reviewed to ensure technical accuracy and consistency with the paper's original contributions.

Acknowledgments

The authors would like to thank all collaborators and reviewers for their valuable feedback and support throughout the development of this work. Vatsavai is supported in part by the AI-LEAF: "AI Institute for Land, Economy, Agriculture and Forestry (AI-LEAF)," and supported by USDA National Institute of Food and Agriculture (NIFA) and the National Science Foundation (NSF) National AI Research Institutes Competitive Award no. 2023-67021-39829. Project website: https://cse.umn.edu/aileaf.

References

[1] Christopher Ardohain and Songlin Fei. 2025. The impacts of training data spatial resolution on deep learning in remote sensing. *Science of Remote Sensing* 11 (June 2025), 100185. https://doi.org/10.1016/j.srs.2024.100185

[2] Favyen Bastani, Piper Wolters, Ritwik Gupta, Joe Ferdinando, and Aniruddha Kembhavi. 2023. SatlasPretrain: A Large-Scale Dataset for Remote Sensing Image Understanding. 16772–16782.

[3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*. 248–255. https://doi.org/10.1109/CVPR.2009.5206848 ISSN: 1063-6919.

[4] Junwu Dong, Yanhui Wang, Yang Yang, Mengqin Yang, and Jun Chen. 2024. MCDNet: Multilevel cloud detection network for remote sensing images based on dual-perspective change-guided and multi-scale feature fusion. *International Journal of Applied Earth Observation and Geoinformation* 129 (May 2024), 103820. https://doi.org/10.1016/j.jag.2024.103820

[5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. https://doi.org/10.48550/arXiv.2010.11929 arXiv:2010.11929 [cs].

[6] Steve Foga, Pat L. Scaramuzza, Song Guo, Zhe Zhu, Ronald D. Dilley, Tim Beckmann, Gail L. Schmidt, John L. Dwyer, M. Joseph Hughes, and Brady Laue. 2017. Cloud detection algorithm comparison and validation for operational Landsat data products. *Remote Sensing of Environment* 194 (June 2017), 379–390. https://doi.org/10.1016/j.rse.2017.03.026

[7] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked Autoencoders Are Scalable Vision Learners. 16000–16009.

[8] Johannes Jakubik, Sujit Roy, and Others. 2023. Foundation Models for Generalist Geospatial Artificial Intelligence. http://arxiv.org/abs/2310.18660 arXiv:2310.18660 [cs].

[9] Lei Ke, Mingqiao Ye, Martin Danelljan, Yifan Liu, Yu-Wing Tai, Chi-Keung Tang, and Fisher Yu. 2023. Segment Anything in High Quality. *Advances in Neural Information Processing Systems* 36 (Dec. 2023), 29914–29934.

[10] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollar, and Ross Girshick. 2023. Segment Anything. 4015–4026.

[11] Kaining Li, Nan Ma, and Lin Sun. 2024. Improving Cloud Detection over Bright Underlying Surfaces Using a Novel MBUNet Framework for Remote Sensing Imagery. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (2024), 1–11. https://doi.org/10.1109/JSTARS.2024.3353262 Conference Name: IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing.

[12] Yuxuan Li, Xiang Li, Yimain Dai, Qibin Hou, Li Liu, Yongxiang Liu, Ming-Ming Cheng, and Jian Yang. 2025. LSKNet: A Foundation Lightweight Backbone for Remote Sensing. *International Journal of Computer Vision* 133, 3 (March 2025), 1410–1431. https://doi.org/10.1007/s11263-024-02247-9

[13] Weixing Liu, Bin Luo, Jun Liu, Han Nie, and Xin Su. 2025. SCTNet: A Shallow CNN–Transformer Network With Statistics-Driven Modules for Cloud Detection. *IEEE Geoscience and Remote Sensing Letters* 22 (2025), 1–5. https://doi.org/10.1109/LGRS.2025.3561004

[14] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. 2021. Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. 9992–10002. https://doi.org/10.1109/ICCV48922.2021.00986 ISSN: 2380-7504.

[15] Ilya Loshchilov and Frank Hutter. 2018. Decoupled Weight Decay Regularization. https://openreview.net/forum?id=Bkg6RiCqY7

[16] Subhajit Paul, Ashutosh Gupta, S. Manthira Moorthi, and Debajyoti Dhar. 2024. C-LiSA: A Hierarchical Hybrid Transformer Model Using Orthogonal Cross Attention for Satellite Image Cloud Segmentation. *IEEE Transactions on Geoscience and Remote Sensing* 62 (2024), 1–14. https://doi.org/10.1109/TGRS.2024.3394929 Conference Name: IEEE Transactions on Geoscience and Remote Sensing.

[17] Libo Wang, Rui Li, Ce Zhang, Shenghui Fang, Chenxi Duan, Xiaoliang Meng, and Peter M. Atkinson. 2022. UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery. *ISPRS Journal of Photogrammetry and Remote Sensing* 190 (Aug. 2022), 196–214. https://doi.org/10.1016/j.isprsjprs.2022.06.008

[18] Xinyi Xu, Wei He, Yu Xia, Hongyan Zhang, Yuwei Wu, Zhen Jiang, and Ting Hu. 2025. TANet: Thin Cloud-Aware Network for Cloud Detection in Optical Remote Sensing Image. *IEEE Transactions on Geoscience and Remote Sensing* 63 (2025), 1–16. https://doi.org/10.1109/TGRS.2025.3540778